

Question:

Write a JAVA program to implement bank functionality in the above scenario and demonstrate

the dynamic polymorphism concept. Note: Create few classes namely Customer, Account, RBI

(Base Class) and few derived classes (SBI, ICICI, PNB etc). Assume and implement required

member variables and functions in each class.

Execution Steps:

Step 1: Program execution starts from main().

Step 2: Scanner object sc is created.

Step 3: User enters RBI.

Step 4: RBI reference variable b is declared.

Step 5: The first if condition is checked.

`name.equalsIgnoreCase("RBI")` is true.

Step 6: RBI object is created.

`b = new RBI();`

Step 7: The `getRate()` method of RBI is called.

Step 8: The method returns 4.0.

Step 9: The interest rate is displayed.

Step 10: Scanner is closed.

Step 11: Program ends.

Algorithm:

Step 1: Start.

Step 2: Create a Scanner object to read input.

Step 3: Enter the bank name.

Step 4: Create a reference variable b of class RBI.

Step 5: Check the entered bank name.

Step 6: If the bank name is RBI, create RBI object.

 If SBI, create SBI object.

 If ICICI, create ICICI object.

 If PNB, create PNB object.

Step 7: If the bank name is invalid,

 display "Invalid bank name"

 and stop the program.

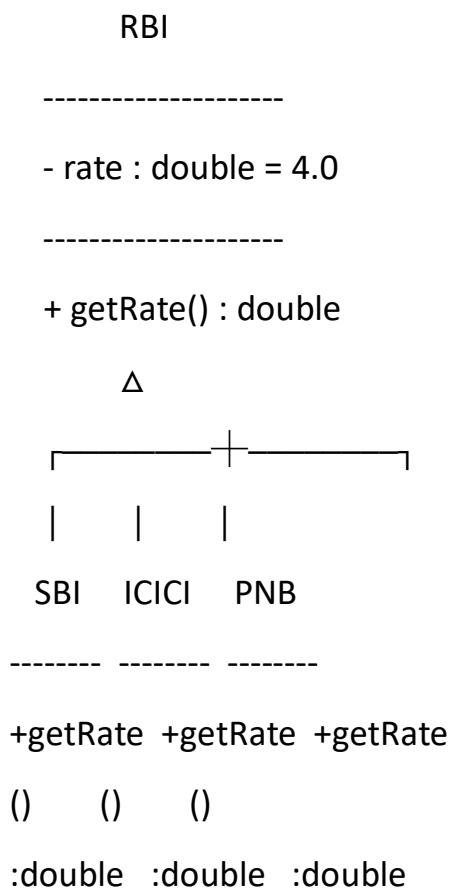
Step 8: Call `getRate()` using reference `b`.

Step 9: Display the rate of interest.

Step 10: Close the Scanner.

Step 11: Stop.

UML Class Diagram:



BankExample

+ main(args:String[])

Code:

```
package javacore;
```

```
import java.util.Scanner;
```

```
class RBI {  
    double rate = 4.0;
```

```
    double getRate() {  
        return rate;  
    }  
}
```

```
class SBI extends RBI {  
    double getRate() {  
        return 7.0;  
    }  
}
```

```
class ICICI extends RBI {  
    double getRate() {  
        return 6.5;
```

```
}  
}
```

```
class PNB extends RBI {  
    double getRate() {  
        return 6.0;  
    }  
}
```

```
public class BankExample {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter the Bank name to find the rate of Interest : ");  
        String name = sc.nextLine();  
  
        RBI b;  
  
        if (name.equalsIgnoreCase("RBI")) {  
            b = new RBI();  
        } else if (name.equalsIgnoreCase("SBI")) {  
            b = new SBI();  
        } else if (name.equalsIgnoreCase("ICICI")) {  
            b = new ICICI();  
        } else if (name.equalsIgnoreCase("PNB")) {
```

```

        b = new PNB();
    } else {
        System.out.println("Invalid bank name");
        sc.close();
        return;
    }

    System.out.println("RBI rate of interest is : " + b.getRate() + "%");

    sc.close();
}
}

```

Enter the Bank name to find the rate of Interest : RBI

RBI rate of interest is : 4.0%

```

"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=61209" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter the Bank name to find the rate of Interest : SBI
RBI rate of interest is : 7.0%

Process finished with exit code 0

```